trait SampleUniformIntBelow

Michael Shoemate

August 11, 2024

This proof resides in **"contrib"** because it has not completed the vetting process.

PR History

- Pull Request #473

This document proves that the implementations of `SampleUniformIntBelow` in `mod.rs` at commit f5bb719 (outdated[1]) satisfy the definition of the `SampleUniformIntBelow` trait.

Definition 0.1. The `SampleUniformIntBelow` trait defines a function `sample_uniform_int_below`.
For any setting of the input parameter `upper`, `sample_uniform_int_below` either

- raises an exception if there is a lack of system entropy,
- returns `out` where `out` is uniformly distributed between $[0, upper)$.

If `trials` is specified, the function will attempt to sample uniformly at random `trials` times, and will raise an exception if it fails to do so. If `trials` is not specified, the function will attempt to sample uniformly at random indefinitely. Setting `trials` causes the function to run in constant time.

There are two `impl`'s (implementations): one for unsigned integers, and one for big integers. To prove correctness of each `impl`, we prove correctness of the implementation of `sample_uniform_int_below`.

Contents

1 `impl` for Unsigned Integers **1**
1.1 Hoare Triple 2
1.2 Proof 2

2 `impl` for Big Integers **2**
2.1 Hoare Triple 3
2.2 Proof 3

1 `impl` for Unsigned Integers

This corresponds to `impl SampleUniformIntBelow for $ty` in Rust. `sample_uniform_int_below` uses rejection sampling. In each round all bits of the integer are filled randomly, drawing an unsigned integer uniformly at random. The algorithm returns the sample, modulo the upper bound, so long as the sample is not one of the final "div" largest integers.

[1] See new changes with `git diff f5bb719..94800cf rust/src/traits/samplers/uniform/mod.rs`

1.1 Hoare Triple

Preconditions

- **User-specified types:**
  - Variable `upper` must be of type `T`
  - Variable `trials` is optional of type `int`, and is non-negative
  - Type `T` is the type the trait is implemented for (one of u8, u16, u32, u64, u128, usize)

Pseudocode

```python
# returns a single bit with some probability of success
def sample_uniform_int_below(upper: int, trials: Optional[int]) -> int:
    found = None
    threshold = T.MAX - T.MAX % upper

    while True:
        if trials == 0:
            if found is None:
                raise ValueError("failed to sample")
            return found
        trials = None if trials is None else trials - 1

        sample = T.sample_uniform_int()
        if sample < threshold and found is None:
            found = sample % upper

        if found is not None and trials is None:
            return found
```

Postcondition

The postcondition is supplied by 0.1.

1.2 Proof

Proof. Assuming that `T.sample_uniform_int()` is correctly implemented, then `v` is a sample between zero and `T.MAX` inclusive, the greatest representable number of type `T`.

You could sample one of `upper` values uniformly at random by rejecting `v` if it is larger than `upper`. That is, only return `v` if `v` is less than `upper`.

It is equivalent to extend the acceptance region, by returning `v % 2` if `v` is less than `upper * 2`, so long as `upper * 2 <= T.MAX`. This reduces the rejection rate, which increases algorithm performance.

There are `T.MAX % upper` remaining elements if you were to extend the acceptance region to the greatest multiple of `upper` that is less than `T.MAX`. Therefore conditioning `v` on being less than `T.MAX - T.MAX % upper` results in `v % upper` being an unbiased, uniformly distributed sample.

When `trials` is specified, the algorithm will attempt to sample uniformly at random `trials` times, and will raise an exception if it fails to do so. Only the first successful sample is kept.
Therefore, for any value of `upper`, the function satisfies the postcondition. □

2 `impl` for Big Integers

This corresponds to `impl SampleUniformIntBelow for UBig` in Rust. This algorithm uses the same algorithm and argument as used for unsigned native integers, but this time the bit depth is dynamically chosen to fill the last byte of a series of bytes long enough to hold `upper`.

2.1 Hoare Triple

Preconditions

- **User-specified types:**
  - Variable `upper` must be of type `UBig`
  - Variable `trials` is optional of type `int`, and is non-negative

Pseudocode

```python
# returns a single bit with some probability of success
def sample_uniform_int_below(upper: int, trials: Optional[int]) -> int:
    byte_len = div_ceil(upper.bit_len(), 8)
    max = Ubig.from_be_bytes([u8.MAX] * byte_len)
    threshold = max - max % upper

    found = None
    buffer = [0] * byte_len
    while True:
        if trials == 0:
            if found is None:
                raise ValueError("failed to sample")
            return found
        trials = None if trials is None else trials - 1

        fill_bytes(buffer)

        sample = UBig.from_be_bytes(buffer)
        if sample < threshold and found is None:
            found = sample % upper

        if found is not None and trials is None:
            return found
```

Postcondition

The postcondition is supplied by 0.1.

2.2 Proof

Proof. `byte_len` is the fewest number of bytes necessary to represent `upper`, which is $ceil(ceil(log_2(upper))/8)$.

This proof follows the same logic as in 1.2, but the constants are generalized. `max` is the largest representable number in `byte_len` bytes, corresponding to `T.MAX`. `v` is an integer sampled uniformly below `max` by randomly filling bits with bernoulli samples. The algorithm terminates when `v` is below the same threshold.

When `trials` is specified, the algorithm will attempt to sample uniformly at random `trials` times, and will raise an exception if it fails to do so. Only the first successful sample is kept.
Therefore, for any value of `upper`, the function satisfies the postcondition. □